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Materials for inflatable carbon fiber mast

Optimal Materials for a Compact Inflatable Composite Carbon Fiber Mast System with Flexible Solar Panels and Anchoring Lander for Lunar Deployment

Quick Reference

Key Findings Table

Subsystem/Component	Recommended Material(s)	Key Properties for Lunar Suitability	Supporting Citations
Inflatable Mast Structure	Silicone-coated Kevlar (aramid) fabric + Shape Memory Polymer (SMP)	High tear resistance, folding endurance, rigidization, low outgassing, thermal stability	1 2
Composite Mast Rigid Elements	High modulus carbon fiber composite (e.g., M55J/RS3) + Protective ceramic coatings	High stiffness, abrasion resistance, thermal cycling durability, low thermal expansion	3 4
Flexible Solar Panel	Flexible CIGS or CdTe thin-film cells on polyimide, with multilayer encapsulation	High efficiency, radiation/UV resistance, mechanical flexibility, thermal cycling stability	5 6 7
Solar Panel Encapsulation/Coating	Polyimide, cerium-doped glass, SiO2, PANDA nanocomposite coatings	UV/electron radiation protection, anti-reflection, dust mitigation	6 8 9
Lander Structure (Body/Legs)	Aluminum honeycomb, carbon fiber composites, basalt fiber composites	Lightweight, impact resistance, vibration damping, thermal cycling resilience	3 10 11 12
Anchoring/Drill System	Alkali-activated regolith geopolymer composites, helical anchor design, PTFE-based seals	High compressive/flexural strength, low reaction force, abrasion resistance, low friction in vacuum	13 14 15
Lubricants/Seals	PTFE-based polymer composites with nano-Al2O3 or nanoclay fillers	Low friction, high wear resistance, low outgassing, vacuum compatibility	15 16
Additive Manufacturing/ISRU	Regolith-epoxy composites, sintered regolith, basalt fiber reinforcement	In-situ fabrication, high strength, thermal stability, resource efficiency	12 17 18

Direct Answer

For your compact inflatable composite carbon fiber mast with integrated communication hardware and a flexible solar panel suitable for lunar deployment, a materials list could include:

1. **Inflatable Mast Structure:**
   - Silicone-coated Kevlar (aramid) fabric with shape memory polymer (SMP) layers for rigidization after inflation. This combination ensures folding durability, high tear resistance, and low outgassing in vacuum, as demonstrated in inflatable lunar habitat studies [1].
2. **Composite Carbon Fiber Components:**
   - High modulus carbon fiber composites (e.g., M55J/RS3) for structural rigidity, enhanced with protective ceramic coatings (e.g., TiO2 or Al2O3 via plasma spraying) to mitigate lunar dust abrasion and thermal cycling effects [3] [4].
3. **Flexible Solar Panel:**
   - Advanced flexible CIGS or CdTe thin-film solar cells on polyimide substrates, with encapsulation technologies such as multilayer UV and anti-reflection coatings (e.g., cerium-doped cover glass) for radiation and dust resistance [5] [6].
4. **Lander Structure and Drill System:**
   - Aluminum honeycomb structures for lightweight impact resistance and rigidity, with carbon fiber-reinforced composites for additional strength.
   - Alkali-activated geopolymer regolith composites for anchoring elements and drill housings, optimized with additives for improved compressive strength and reduced porosity [11] [13].
5. **Lubricants and Seals:**
   - PTFE-based polymer composites with nano-Al2O3 or nanoclay fillers for moving parts such as the drill system, minimizing friction and wear under vacuum, and ensuring chemical compatibility and low outgassing [15].

Study Scope

- **Time Period:** 2018–2024 (emphasis on post-2020 advancements)
- **Disciplines:** Materials science, aerospace engineering, lunar construction, additive manufacturing, photovoltaics
- **Methods:** Meta-analysis of experimental studies, simulation, in-situ lunar analog testing, advanced manufacturing trials

Assumptions & Limitations

- Most data are derived from lunar simulant studies and analog environments; direct long-term lunar surface validation is limited.
- Integration of multiple advanced materials in a single deployable system is still an emerging field; system-level interactions under simultaneous lunar stressors require further study.
- Some material recommendations (e.g., self-healing coatings, advanced encapsulants) are based on recent laboratory demonstrations and may need further qualification for lunar deployment.

Suggested Further Research

- Long-term, integrated system testing of inflatable composite masts with embedded solar panels and communication hardware under simulated lunar conditions (thermal, vacuum, dust, radiation).
- In-situ validation of regolith-based composites and geopolymer anchoring systems on the lunar surface.
- Development and qualification of multifunctional coatings (self-healing, dust-repellent, radiation-shielding) for both structural and photovoltaic components.
- Optimization of additive manufacturing processes for hybrid regolith-polymer and regolith-fiber composites using ISRU.

1. Introduction

Overview of System Concept

The proposed system is a compact, deployable lunar surface platform comprising:

- An **inflatable mast** constructed from advanced composite materials, integrating communication hardware.
- **Flexible solar panels** attached to the mast, providing power generation.
- All components are stowed within a **cylindrical lander** equipped with three legs and a **low reaction force drill system** for anchoring and rigid support on the lunar surface.

Material selection is critical for ensuring durability, functionality, and survivability in the Moon's extreme environment, which includes vacuum, wide temperature swings, intense radiation, and abrasive dust [1] [10] [19] [20].

2. Lunar Environmental Compatibility

Material Performance Under Lunar Conditions

- **Vacuum:** Materials must exhibit low outgassing and maintain mechanical integrity without atmospheric support.
- **Temperature Extremes:** Lunar day-night cycles range from -170°C to +130°C, requiring materials with low thermal expansion and high thermal cycling resistance.

- **Radiation:** High-energy solar and cosmic radiation necessitate radiation-hardened materials and coatings.
- **Abrasive Dust:** Lunar regolith is highly abrasive and electrostatically charged, demanding wear-resistant surfaces and dust mitigation strategies [13] [21] [22].

Top-performing materials include:

- Geopolymers and sintered regolith for construction [23] [24].
- High-performance ceramics (e.g., 8YSZ, Ti6Al4V+hBN) for coatings [21] [24].
- Advanced polymers with non-stick and thermo-mechanical stability [25] [26].

Radiation-Resistant Coatings and Dust Mitigation

- **Self-healing ceramics, high-entropy ceramics, and MAX phases** provide microstructural stability and crack resistance under radiation [27].
- **Ti/2 vol% hBN coatings** (plasma sprayed) offer wear resistance, radiation-induced hardening, and neutron shielding [21].
- **Aluminum oxide and vanadium dioxide coatings** enhance abrasion and thermal resistance [28] [29].
- **Superhydrophobic polyimide-based coatings** and **electrodynamic dust shields** reduce dust adhesion and improve longevity under thermal cycling [30] [31].

Mechanical Property Evolution Under Lunar Stressors

- **Ti6Al4V+hBN coatings** show increased hardness and reduced wear volume after lunar thermal cycling and radiation exposure [21] [32].
- **Amorphous alloys** and advanced composites maintain mechanical properties under repeated lunar day-night cycles [33].

Synthesis:
Material selection for lunar systems must prioritize resistance to vacuum-induced outgassing, thermal cycling, radiation, and abrasive dust. Advanced coatings and hybrid composites are essential for long-term durability [13] [21] [22].

3. Materials for Inflatable Mast and Composite Carbon Fiber Components

Silicone-Coated Kevlar and Shape Memory Polymers

- **Silicone-coated Kevlar (aramid) fabric** provides high tear resistance, folding endurance (retains 29% strength after 50,000 cycles), and operational temperature range from -62°C to +230°C [1] [34].
- **Shape memory polymers (SMPs)** enable rigidization post-inflation, enhancing structural stability and vibration control [2].

- **Nanoparticle coatings** (SiC, TiO2) further improve thermal management and raise glass transition temperature [35].
- **Ultra-low outgassing silicones** are essential for vacuum compatibility; pre/post-heating treatments reduce outgassing to <0.2% TML [36] [37].

High Modulus Carbon Fiber Composites

- **M55J/RS3 and similar composites** offer high stiffness, low thermal expansion, and abrasion resistance after lunar dust exposure and thermal cycling [3] [4].
- **Protective ceramic coatings** (e.g., TiO2, Al2O3) applied via plasma spraying enhance surface durability [22].
- **Surface treatments** (plasma, laser) improve fiber-matrix bonding and abrasion resistance [38] [39].

Regolith-Based Composite Additives

- **Graphene, carbon nanotubes, and basalt fibers** added to regolith composites significantly increase compressive and flexural strength (up to 67 MPa) [12] [40] [41].
- **3D printing and additive manufacturing** enable integration of these additives for automated, in-situ fabrication [18] [42].

Fatigue, Wrinkle, and Stress Mitigation in Inflatable Membranes

- **Fatigue and folding endurance** are limited by micro-cracking and fiber/matrix debonding after thermal cycling; design safety margins of 20% are recommended [43] [44].
- **Wrinkle mitigation** strategies include shear-compliant borders, web-cable girding, and shape optimization to maintain membrane tension and reduce stress concentrations [45] [46] [47].

Synthesis:
A hybrid inflatable mast using silicone-coated Kevlar with SMPs, reinforced by high modulus carbon fiber composites and regolith-based additives, offers optimal mechanical and environmental performance for lunar deployment [1] [3] [4].

4. Flexible Solar Panel Materials for Lunar Deployment

High-Efficiency Flexible Solar Cell Technologies

- **Flexible CIGS (Cu(In,Ga)Se2) solar cells** on polyimide substrates achieve >20% efficiency and exhibit strong radiation hardness [48] [49].
- **Flexible CdTe solar cells** reach up to 11.4% efficiency, with lightweight and flexible form factors [50].
- **Flexible perovskite solar cells** show high efficiency (~12%), excellent radiation tolerance, and mechanical flexibility (retain >90% efficiency after 5,000 bending cycles) [7] [51] [52].
- **Flexible GaAs thin-film modules** offer high specific power (>750 W/kg) and reliability [5].

Radiation and UV Protection: Encapsulation and Coatings

- **Polyimide, cerium-doped glass, and SiO2 coatings** provide UV and electron radiation protection, anti-reflection, and dust mitigation [6] [8] [9].
- **PANDA nanocomposite coatings** enhance UV stability and convert absorbed UV into visible light, boosting photovoltaic efficiency [8].
- **Multilayer encapsulation** strategies (e.g., transparent conductive oxides, anti-reflection layers) are critical for long-term durability [53] [54].

Mechanical and Thermal Stability of Flexible Solar Cells

- **Flexible perovskite and CIGS cells** maintain high efficiency and mechanical integrity after extensive thermal cycling (-80°C to +80°C) and vacuum exposure [52] [55] [56].
- **Critical bending radius** for flexible perovskite cells is ~0.5–1.0 mm; advanced buffer layers and substrate optimization improve endurance [57] [58].
- **CdTe and CIGS trade-offs:** CdTe offers high absorption but contains toxic cadmium; CIGS is more environmentally benign but relies on scarce elements [59] [60].

Synthesis:
Flexible CIGS or CdTe solar cells on polyimide, with advanced encapsulation (cerium-doped glass, SiO2, PANDA coatings), provide the best balance of efficiency, radiation resistance, and mechanical durability for lunar surface power [5] [6] [7].

5. Materials for Low Reaction Force Drill System and Lander Legs

Aluminum Honeycomb and Composite Structures

- **Aluminum honeycomb** offers high strength-to-weight ratio, impact resistance, and vibration damping, ideal for lander legs and structural panels [10] [11].
- **Carbon fiber and basalt fiber composites** provide additional rigidity and thermal cycling resilience [3] [12].
- **Origami-inspired and multilayer honeycomb designs** enhance energy dissipation and thermal management [61] [62].

Regolith-Based and Alkali-Activated Geopolymer Materials

- **Alkali-activated geopolymers** (using sodium silicate) achieve compressive strengths up to 44 MPa and maintain microstructural stability after thermal cycling [13] [63].
- **Basalt fiber reinforcement** further improves flexural and tensile properties [12] [63].
- **Curing at elevated temperatures** (e.g., 90°C) optimizes strength and durability [64].

Low Binder-Content Regolith Composites and Helical Anchors

- **Low binder-content regolith composites** (4–15% epoxy or PEKK) achieve flexural strengths of 12–30 MPa, sufficient for structural components [65] [66].
- **Helical anchor designs** maximize lateral resistance and penetration depth, critical for stable anchoring in compacted regolith [14] [67].

Optimizing Drill System Materials and Cutting Geometries

- **PTFE-based polymer composites** with nano-Al2O3 or nanoclay fillers minimize friction and wear in vacuum, ideal for seals and lubricants in the drill system [15] [16].
- **Cutting edge geometry** (increased rake angle, percussive frequency) reduces reaction force and improves drilling efficiency [68].

Synthesis:
A combination of aluminum honeycomb, carbon/basalt fiber composites, alkali-activated regolith geopolymers, and optimized PTFE-based seals provides a lightweight, robust, and reliable lander structure and anchoring system for lunar deployment [10] [11] [13] [15].

6. Optimal Material Synthesis for Lunar Composite Systems

Integrated Composite System Design

- **Hybrid structures** combine silicone-coated Kevlar/SMP inflatable membranes, high modulus carbon fiber composites, and regolith-based additives for optimal weight, durability, and environmental resilience [40] [69] [70].
- **Protective coatings** (ceramic, self-healing, superhydrophobic) are applied to all exposed surfaces for dust and radiation mitigation [4] [30].

Polymer Matrix Composites and Dust Resistance

- **PTFE-based composites** for seals and moving parts, with nano-fillers for enhanced wear resistance [15] [16].
- **Regolith-resin composites** (epoxy, cyanate ester, PEKK) for structural parts, offering low thermal expansion and high mechanical strength [66] [71].
- **Basalt fiber reinforcement** for improved durability and thermal stability [12] [72].

Additive Manufacturing and In-Situ Resource Utilization

- **Selective laser sintering, material extrusion, and microwave sintering** enable in-situ fabrication of regolith-based composites, reducing Earth-supplied mass [17] [73].
- **Hybrid regolith-polymer and regolith-fiber composites** can be tailored for specific mechanical and thermal requirements [12] [18].

Inorganic Polymer Systems and Advanced Binders

- **Geopolymers and cyanate ester binders** synthesized from lunar elements (Si, Al, O) serve as adhesives and matrix materials, offering high compressive strength and thermal stability [74] [75].

- **Graphene oxide and nano-SiO2 additives** further enhance mechanical properties and microstructural refinement [76].

Synthesis:
A system-level approach integrating advanced composites, protective coatings, and ISRU-enabled manufacturing yields a multifunctional, lightweight, and durable lunar deployment platform [4] [40] [69] [70] [77] [78] [79].

7. Conclusion

Summary of Optimal Materials and System Integration

- **Inflatable Mast:** Silicone-coated Kevlar with SMPs for rigidization, nanoparticle coatings for thermal management, and design features to mitigate wrinkles and stress concentrations.

- **Composite Carbon Fiber Components:** High modulus carbon fiber (M55J/RS3) with ceramic coatings for rigidity, abrasion resistance, and low thermal expansion.

- **Flexible Solar Panels:** Flexible CIGS or CdTe cells on polyimide, encapsulated with multilayer UV/radiation-resistant coatings (cerium-doped glass, SiO2, PANDA).

- **Lander Structure and Drill System:** Aluminum honeycomb and carbon/basalt fiber composites for lightweight strength; alkali-activated regolith geopolymers and low binder-content composites for anchoring; PTFE-based seals and lubricants for moving parts.

- **Additive Manufacturing/ISRU:** Regolith-based composites and basalt fiber reinforcement for in-situ fabrication and resource efficiency.

Collectively, these materials address the unique challenges of the lunar environment—vacuum, temperature extremes, radiation, and abrasive dust—while enabling a compact, deployable, and robust lunar surface system.

Future Research Directions

- **Integrated system testing** under simulated lunar conditions to validate long-term performance and interactions of combined materials.

- **In-situ lunar trials** for regolith-based composites, anchoring systems, and inflatable structures.

- **Advanced coatings** development for multifunctional protection (self-healing, dust-repellent, radiation-shielding).

- **Optimization of additive manufacturing** for hybrid composites using ISRU, including digital twin and AI-assisted design for rapid iteration and deployment.

In summary, the recommended materials and synthesis strategies provide a robust foundation for developing a compact inflatable composite carbon fiber mast system with flexible solar panels and a rigid, anchored lander, optimized for the harsh lunar environment and future lunar surface operations [1] [10] [19] [20] [40].

References

1. Rigidization capacity and folding behavior of variable stiffness composite skin for inflatable lunar habitats Wang, Q., Feng, P., Wu, B., (...), Jansen, K. Composites Part A: Applied Science and Manufacturing, 2025 https://www.scopus.com/pages/publications/105011185190?origin=scopusAI

2. A novel rigidizable inflatable lunar habitation system: design concept and material characterization Wang, Q., Feng, P., Jansen, K., Bao, C. Materials and Design, 2024 https://www.scopus.com/pages/publications/85203632454?origin=scopusAI

3. Evaluation of the resistance of composite materials to lunar dust abrasion Boucher, M.L., Potvin, M.-J. 61st International Astronautical Congress 2010, IAC 2010, 2010 https://www.scopus.com/pages/publications/79959457663?origin=scopusAI

4. Resistance of thermally cycled composites to lunar dust abrasion Martin, F., Potvin, M.-J. 26th Annual Technical Conference of the American Society for Composites 2011 and the 2nd Joint US-Canada Conference on Composites, 2011 https://www.scopus.com/pages/publications/84865619513?origin=scopusAI

5. High-Efficiency Flexible GaAs Solar Modules Used for Unmanned Aerial Vehicles and Stratospheric Airships Yang, J.K., Jiao, X.Y., Yang, Y., (...), Chen, Z.C. Lecture Notes in Electrical Engineering, 2020 https://www.scopus.com/pages/publications/85087781247?origin=scopusAI

6. Development and testing of coatings for orbital space radiation environments Pellicori, S.F., Martinez, C.L., Hausgen, P., Wilt, D. Applied Optics, 2014 https://www.scopus.com/pages/publications/84942370547?origin=scopusAI

7. Efficient minority carrier detrapping mediating the radiation hardness of triple-cation perovskite solar cells under proton irradiation Lang, F., Jošt, M., Bundesmann, J., (...), Nickel, N.H. Energy and Environmental Science, 2019 https://www.scopus.com/pages/publications/85065961376?origin=scopusAI

8. UV Stable Thin-film Down-converting Polymer Coating for Sail-embedded PV Power Sarkisov, S., Patel, D., Pedoseyev, A. Conference Record of the IEEE Photovoltaic Specialists Conference, 2024 https://www.scopus.com/pages/publications/85211613164?origin=scopusAI

9. Atomic oxygen resistant coatings on Kapton Ritchie, I., Gjerde, H.B. Surface and Coatings Technology, 1989 https://www.scopus.com/pages/publications/0024911669?origin=scopusAI

10. Semi-actively controlled landing legs for a spacecraft Hashimoto, T., Maeda, T., Otsuki, M., Mashimo, T. Proceedings of the International Astronautical Congress, IAC, 2016 https://www.scopus.com/pages/publications/85016508964?origin=scopusAI

11. 3-D woven honeycomb structures and their composites Tripathi, L., Singh, O., Behera, B.K. Textile Progress, 2024 https://www.scopus.com/pages/publications/85189519448?origin=scopusAI

12. Basalt fiber composites for the robotic fabrication of a lunar habitat Cheibas, I., Arnhof, M., Rich, B. Proceedings of the International Astronautical Congress, IAC, 2022 https://www.scopus.com/pages/publications/85167571136?origin=scopusAI

13. Combined effect of curing temperature, curing period and alkaline concentration on the mechanical properties of fly ash-based geopolymer Sajan, P., Jiang, T., Lau, C., (...), Ng, K. Cleaner Materials, 2021 https://www.scopus.com/pages/publications/85118544345?origin=scopusAI

14. Helical anchors: Lateral resistance for shaping the lunar surface Crabtree, J., Klosky, J.L., Gash, R., Miller, D. Earth and Space Conference 2008: Proceedings of the 11th Aerospace Division International Conference on

Engineering, Science, Construction, and Operations in Challenging Environments, 2008 https://www.scopus.com/pages/publications/70349383851?origin=scopusAI

15. Tribological Behavior of Nano-Al₂O₃-Reinforced PPS-PTFE Composites Wang, Y., Gong, J., Yang, D., (...), Wang, H. Tribology Transactions, 2014 https://www.scopus.com/pages/publications/84896518627?origin=scopusAI

16. Tribological properties of PTFE composites in lunar extreme environment Matsumoto, K., Suzuki, M., Takada, S. Proceedings of the International Astronautical Congress, IAC, 2012 https://www.scopus.com/pages/publications/84883520488?origin=scopusAI

17. Additive manufacturing of lunar regolith: A review Sun, W., Dang, F., Zhou, Y., (...), Ding, L. Additive Manufacturing Frontiers, 2025 https://www.scopus.com/pages/publications/105011666712?origin=scopusAI

18. 3D printing of regolith-based epoxy composites with excellent temperature resistance and mechanical strength Bao, C., Wang, Y., Pearce, G., (...), Mushtaq, R.T. Acta Astronautica, 2025 https://www.scopus.com/pages/publications/85216870420?origin=scopusAI

19. Mechanical Properties of Lunar Concrete: Experimental and Simulation Perspectives Pilehvar, S., Kassem, A., Osman, E.H., (...), Kioumarsi, M. Lecture Notes in Civil Engineering, 2026 https://www.scopus.com/pages/publications/105045449182?origin=scopusAI

20. Hybrid multilayer radiation shielding for lunar habitats: material integration, encapsulation strategy, and optimization against GCR and SPE Owaisi, G.M., Yang, Q., Bartoloni, A., Yuan, S. Advances in Space Research, 2026 https://www.scopus.com/pages/publications/105047441012?origin=scopusAI

21. Wear and neutron shielding resilience of titanium-hexagonal boron nitride coatings against extreme lunar radiation and thermal cycles Sukumaran, A.K., Orikasa, K., Rengifo, S., (...), Agarwal, A. Surface and Coatings Technology, 2024 https://www.scopus.com/pages/publications/85201128461?origin=scopusAI

22. Morphological and mechanical studies of Al₂O₃-Na₂SiO₃ as a skin barrier coated with TiO₂ for carbon fiber reinforced composite materials Widyastuti, Hardiyanti, S., Muhaqqi Al Haq, W.S., (...), Wahyuono, R.A. RSC Advances, 2024 https://www.scopus.com/pages/publications/85188534584?origin=scopusAI

23. Optimizing cementitious mortars for lunar construction: a multi-criteria decision-making approach Karbala, M.M., Goudarzi, S., Nejati, H.R., Abedini, M. Journal of Sustainable Cement-Based Materials, 2026 https://www.scopus.com/pages/publications/105017391726?origin=scopusAI

24. Investigating the Resilience of 8 mol% Yttria-Stabilized Zirconia Coatings for Lunar Environments Pujols, A.T., Stein, Z., Wohl, C., (...), Raghavan, S. AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2025, 2025 https://www.scopus.com/pages/publications/85219607023?origin=scopusAI

25. Dust Mitigation Strategies Enabling Moon Exploration Missions Saccone, G., Favaloro, N. Aerotecnica Missili and Spazio, 2025 https://www.scopus.com/pages/publications/105030533496?origin=scopusAI

26. DUST mitigation technology for lunar exploration and colonization: existing and future perspectives Saccone, G., Favaloro, N. Materials Research Proceedings, 2023 https://www.scopus.com/pages/publications/85185790555?origin=scopusAI

27. Radiation-tolerant coatings for extreme environments: Advancements in ceramics, composites, and coating techniques for nuclear and aerospace applications Kanu, N.J., Gupta, E., Singh, G.K., Vates, U.K. Materials Today Communications, 2026 https://www.scopus.com/pages/publications/105044307158?origin=scopusAI

28. Investigating the Durability of Aluminum Oxide Coatings for Lunar Dust Mitigation Latorre-Suarez, P., Gucwa, K., Swaby, O., (...), Raghavan, S. Proceedings of the International Astronautical Congress, IAC, 2022 https://www.scopus.com/pages/publications/85167601510?origin=scopusAI

29. Radiation resilient epitaxial VO₂ thin films for thermal and optoelectronic integration in spaceborne electronic subsystems Singh, S., Anand, J.K., Saini, S., (...), Singh, R. Applied Materials Today, 2026 https://www.scopus.com/pages/publications/105026450560?origin=scopusAI

30. Conductive polyimide coatings for passive lunar dust mitigation Eyckens, D.J., Yang, W., Holmes, S., (...), Simons, R. Applied Surface Science, 2026 https://www.scopus.com/pages/publications/105027460403?origin=scopusAI

31. Performance of electrodynamic dust shields with functional surface coatings for lunar dust mitigation Scifoni, L., Isidori Pacelli, F., Schaible, M.J., (...), Romero-Calvo, Á. Acta Astronautica, 2026 https://www.scopus.com/pages/publications/105016098721?origin=scopusAI

32. Erosion behavior of Ti-hBN multifunctional coatings in a custom-made planetary test rig at extreme lunar temperatures Sukumaran, A.K., Rengifo, S., Aguiar, B., (...), Agarwal, A. Tribology International, 2025 https://www.scopus.com/pages/publications/85207792955?origin=scopusAI

33. Property evolution and service life prediction of novel metallic materials for future lunar bases Meng, S.-K., Zhu, R.-S., Zheng, H.-L., (...), Wang, Z. Rare Metals, 2024 https://www.scopus.com/pages/publications/85195599448?origin=scopusAI

34. Silicone-coated Kevlar sheeting New Metals Chemicals Ltd High Performance Textiles, 1991 https://www.scopus.com/pages/publications/0025958177?origin=scopusAI

35. Enhancement of thermal properties of Kevlar 29 coated by SiC and TiO₂ nanoparticles and their binding energy analysis Akibul Islam, M., Chowdhury, M.A., Arefin Kowser, M., (...), Ramjan Ali, M. Arabian Journal of Chemistry, 2022 https://www.scopus.com/pages/publications/85131060938?origin=scopusAI

36. Characteristics of elastomer seals exposed to space environments Daniels, C.C., De Groh III, H.C., Dunlap Jr., P.H., (...), Waters, D.L. Collection of Technical Papers - 43rd AIAA/ASME/SAE/ASEE Joint Propulsion Conference, 2007 https://www.scopus.com/pages/publications/36749043686?origin=scopusAI

37. Preparation and application research of composites with low vacuum outgassing and excellent electromagnetic sealing performance Mao, H., Zou, H., Liu, W., (...), Xing, B. E-Polymers, 2024 https://www.scopus.com/pages/publications/85189664897?origin=scopusAI

38. Recent Advances in Interfacial Strengthening and Matrix Toughening of Carbon-Fiber-Reinforced Polymer Composites Wang, Y., Yu, H., Wang, L., (...), Xie, Q. Coatings, 2026 https://www.scopus.com/pages/publications/105051828164?origin=scopusAI

39. Characterization of CFRP interface properties with varying fiber surface roughness using AFM measurements and finite element modeling Hoang, Q.N., Lee, S., Lee, S., Park, H. Composites Part B: Engineering, 2026 https://www.scopus.com/pages/publications/105015586120?origin=scopusAI

40. Regolith-based magnesium oxychloride composites doped by graphene: Novel high-performance building materials for lunar constructions Lauermannová, A.-M., Faltysová, I., Lojka, M., (...), Jankovský, O. FlatChem, 2021 https://www.scopus.com/pages/publications/85100642986?origin=scopusAI

41. A COMPOSITIONAL STUDY OF REGOLITH COMPOSITES WITH CARBON NANOTUBE ADDITIVES FOR EXTRATERRESTRIAL CONSTRUCTION Hoe, A.J., Lin, W., Wang, Y. Proceedings of ASME 2023 Aerospace Structures, Structural Dynamics, and Materials Conference, SSDM 2023, 2023 https://www.scopus.com/pages/publications/85176783048?origin=scopusAI

42. Lunar regolith - AlSi10Mg composite fabricated by selective laser melting Liao, H., Zhu, J., Chang, S., (...), Zhu, H. Vacuum, 2021 https://www.scopus.com/pages/publications/85100619686?origin=scopusAI

43. Effect of thermal cycling on the anisotropic and loading–unloading mechanical behavior of plain-woven aramid-fiber/epoxy composites Liu, Q., Wu, W., Zhou, Y., (...), Lu, T.J. Composite Structures, 2026 https://www.scopus.com/pages/publications/105034579320?origin=scopusAI

44. Experimental study of large scale model composite materials under thermal fatigue Biernacki, K., Szyszkowski, W., Yannacopoulos, S. Composites Part A: Applied Science and Manufacturing, 1999 https://www.scopus.com/pages/publications/0032662116?origin=scopusAI

45. Design parameter effects for wrinkle reduction in membrane space structures Sakamoto, H., Park, K.C. Collection of Technical Papers - AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference, 2005 https://www.scopus.com/pages/publications/28844492219?origin=scopusAI

46. Influence parameter analysis and wrinkling control of space membrane structures Wang, C.G., Mao, L.N., Du, X.W., He, X.D. Mechanics of Advanced Materials and Structures, 2010 https://www.scopus.com/pages/publications/76149098852?origin=scopusAI

47. Stress-based structural optimization for wrinkle suppression of membrane structures via ANCF-SRM Ji, Y., Sun, J., Jin, D. Computer Methods in Applied Mechanics and Engineering, 2026 https://www.scopus.com/pages/publications/105042671586?origin=scopusAI

48. High efficiency flexible Cu(In,Ga)Se₂ solar cells Reinhard, P., Chirila, A., Pianezzi, F., (...), Tiwari, A.N. Proceedings of the 20th International Workshop on Active-Matrix Flatpanel Displays and Devices: TFT Technologies and FPD Materials, AM-FPD 2013, 2013 https://www.scopus.com/pages/publications/84887828791?origin=scopusAI

49. Prototype development - Flexible Cu(In,Ga)Se₂ thin film solar cells for space applications Neisser, A., Kaufmann, C.A., Klenk, R., (...), Schock, H.-W. Conference Record of the IEEE Photovoltaic Specialists Conference, 2005 https://www.scopus.com/pages/publications/27944494326?origin=scopusAI

50. High-efficiency flexible CdTe solar cells on polymer substrates Romeo, A., Khrypunov, G., Kurdesau, F., (...), Tiwari, A.N. Solar Energy Materials and Solar Cells, 2006 https://www.scopus.com/pages/publications/33748331192?origin=scopusAI

51. Dependence of ITO-Coated Flexible Substrates in the Performance and Bending Durability of Perovskite Solar Cells Pandey, M., Wang, Z., Kapil, G., (...), Hayase, S. Advanced Engineering Materials, 2019 https://www.scopus.com/pages/publications/85066914547?origin=scopusAI

52. Enhanced Performance of Flexible Perovskite Solar Cells Enabled by ATB Passivator Yang, Y., Fan, J., Jiang, T., (...), Hao, X. Energy and Fuels, 2024 https://www.scopus.com/pages/publications/85199196167?origin=scopusAI

53. Novel environmental protection for multi-junction solar cells Martinez, C.L., Pellicori, S.F. 2015 IEEE 42nd Photovoltaic Specialist Conference, PVSC 2015, 2015 https://www.scopus.com/pages/publications/84961615744?origin=scopusAI

54. Test results of radiation resistant coatings integrated with flexible substrate materials and active solar cells Martinez, C.L., Pellicori, S.F. 2017 IEEE 44th Photovoltaic Specialist Conference, PVSC 2017, 2017 https://www.scopus.com/pages/publications/85048489300?origin=scopusAI

55. Redox-Buffering PEAI-CeO_x Passivation Suppresses Thermo-Mechanical Degradation in Perovskite Solar Cells Under Rapid Thermal Cycling Asare, G.K., Lee, M., Adu, J.S., (...), Park, H.H. Advanced Energy Materials, 2026 https://www.scopus.com/pages/publications/105045960902?origin=scopusAI

56. Towards space compatible perovskite solar cells: guidelines for thermal shock resilience and near space balloon testing Lee, M., Asare, G.K., Richards, T.H., (...), Yun, J.S. Energy and Environmental Science, 2016 https://www.scopus.com/pages/publications/105030979285?origin=scopusAI

57. Nanomechanical Approach for Flexibility of Organic-Inorganic Hybrid Perovskite Solar Cells Ahn, S.-M., Jung, E.D., Kim, S.-H., (...), Kim, J.-Y. Nano Letters, 2019 https://www.scopus.com/pages/publications/85066904185?origin=scopusAI

58. High Thermal Conductivity of Liquid Crystal Elastomer for Stress-Less Flexible Perovskite Solar Cells Ma, Y., You, J., Zhang, L., (...), Liu, S. Advanced Functional Materials, 2024 https://www.scopus.com/pages/publications/85196406263?origin=scopusAI

59. CdTe Photovoltaics in Superstrate Geometry-Back-Contact Materials: A Review Narayanan, V., Ks, R. ACS Applied Energy Materials, 2023 https://www.scopus.com/pages/publications/85172703009?origin=scopusAI

60. Flexible CIGS, CdTe and a-Si:H based thin film solar cells: A review Ramanujam, J., Bishop, D.M., Todorov, T.K., (...), Romeo, A. Progress in Materials Science, 2020 https://www.scopus.com/pages/publications/85076851718?origin=scopusAI

61. Cushioning performance of origami negative poisson's ratio honeycomb steel structure Zhou, Y., Jiang, D., Wang, L., (...), Jia, L.-J. Thin-Walled Structures, 2024 https://www.scopus.com/pages/publications/85200637844?origin=scopusAI

62. Bioinspired multilayer barriers of 3D-printed lunar regolith simulant-based geopolymers for mechanical and thermal protection Ma, S., Zhang, F., Fu, S., (...), Zhou, Y. Additive Manufacturing, 2026 https://www.scopus.com/pages/publications/105030139572?origin=scopusAI

63. Mechanical and microstructural performance of LHS-1E-based geopolymer concrete subjected to thermal curing conditions Faysal, R.M., Smith, S.T., Tafsirojjaman, T. Construction and Building Materials, 2026 https://www.scopus.com/pages/publications/105051430105?origin=scopusAI

64. Reactive components and alkali leaching behavior of lunar regolith simulant for geopolymer production Xue, G., Wang, R., Sheng, P., Qiao, G. Construction and Building Materials, 2025 https://www.scopus.com/pages/publications/105024313021?origin=scopusAI

65. Impact formation of ultralow-binder-content composite “lunar cement” Oh, K., Yi, H., Chen, T., (...), Qiao, Y. CEAS Space Journal, 2021 https://www.scopus.com/pages/publications/85090121647?origin=scopusAI

66. Ultralow-binder-content thermoplastic composites based on lunar soil simulant Oh, K., Chen, T., Kou, R., (...), Qiao, Y. Advances in Space Research, 2020 https://www.scopus.com/pages/publications/85089825596?origin=scopusAI

67. Experimental Study on Compressive Capacity Behavior of Helical Anchors in Aeolian Sand and Optimization of Design Methods Chen, Q., Liu, W., Li, L., (...), Guo, Y. Buildings, 2025 https://www.scopus.com/pages/publications/105011604430?origin=scopusAI

68. Research on Low-Force Cutting Load Characteristics for Lunar Rock Simulant Wang, Y., Li, P., Xu, B., Liu, J. Proceedings of SPIE - The International Society for Optical Engineering, 2022 https://www.scopus.com/pages/publications/85128270647?origin=scopusAI

69. Structural materials from lunar simulants through thermal liquefaction Desai, Chandra S., Girdner, Kirsten null, 1992 https://www.scopus.com/pages/publications/0026727528?origin=scopusAI

70. Composite materials for structures on planetary surfaces Radford, Donald W., Sadeh, Willy Z., Cheng, Boyle C. null, 1992 https://www.scopus.com/pages/publications/0027100486?origin=scopusAI

71. Behaviour of regolith-resin-composites subjected to lunar analogue environments Hossain, M.A., Smith, S.T., Tafsirojjaman, T. Journal of Building Engineering, 2026 https://www.scopus.com/pages/publications/105042672800?origin=scopusAI

72. Durability of Basalt Fiber Fabric Under Simulated Lunar Extreme Thermal and UV Conditions: A Comparative Study with Natural Basalt Rock Qin, J., Cai, M., Wang, Y., (...), Xu, W. Polymers, 2026 https://www.scopus.com/pages/publications/105047160562?origin=scopusAI

73. Solar-Driven Additive Sintering of Lunar Regolith Simulant: A Proof-of-Concept for Off-Earth Manufacturing Sreekumar, S., Iversen, M., Afanador, N., (...), Valdivia, A. AIAA Science and Technology

Forum and Exposition, AIAA SciTech Forum 2026, 2026
https://www.scopus.com/pages/publications/105031182488?origin=scopusAI

74. LUNAR BUILDING MATERIALS - SOME CONSIDERATIONS ON THE USE OF INORGANIC POLYMERS. Lee, Stuart M. NASA Special Publications, 1979
https://www.scopus.com/pages/publications/0018738475?origin=scopusAI

75. Cyanate Ester–Lunar Regolith Composites for In Situ Fabrication of Structural Electronics on the Moon Li, G., Alasahin, B.M., Mirotznik, M., Opila, R.L. Electronics (Switzerland), 2026
https://www.scopus.com/pages/publications/105040130395?origin=scopusAI

76. Preparation and properties of graphene oxide and nano-SiOi reinforced lunar regolith simulant geopolymer Zhang, B.-B., Ma, Q.-Y. Yantu Lixue/Rock and Soil Mechanics, 2026
https://www.scopus.com/pages/publications/105037386467?origin=scopusAI

77. Development, testing and modeling of ceramic composites for lunar applications Toth, Janos C., Desai, Chandra S. Proceedings of the 4th International Conference on Engineering, Construction, and Operations in Space, 1994 https://www.scopus.com/pages/publications/0028253220?origin=scopusAI

78. Preparation and characterization of elevated and cryogenic temperature-resistant regolith-based epoxy resin composites Bao, C., Wang, Y., Tahir Mushtaq, R., (...), Liu, M. Construction and Building Materials, 2023
https://www.scopus.com/pages/publications/85156224333?origin=scopusAI

79. Lunar base construction materials and technologies: A review Li, G., Zhang, J., Ke, S., (...), Zhao, K. Acta Astronautica, 2026 https://www.scopus.com/pages/publications/105043137060?origin=scopusAI